I-GUIDE

Institute for Geospatial Understanding through an Integrative Discovery Environment

Convergence Curriculum for Geospatial Data Science

Eric Shook & Jinzhe Wang
University of Minnesota

2026 AAG Annual Meeting

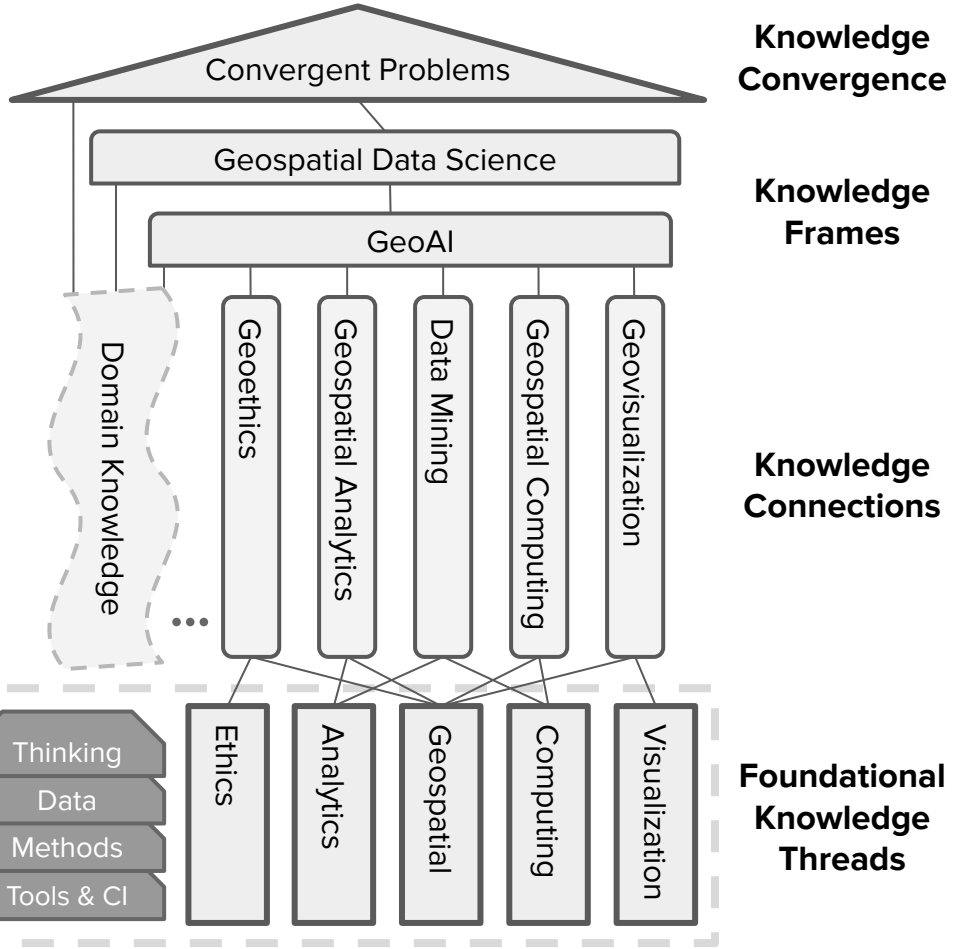
Award #: 2118329

March 20, 2026

Convergence Curriculum for Geospatial Data Science

- Support multiple learner pathways
- Structure building knowledge, skills, and competencies infused with advanced technologies

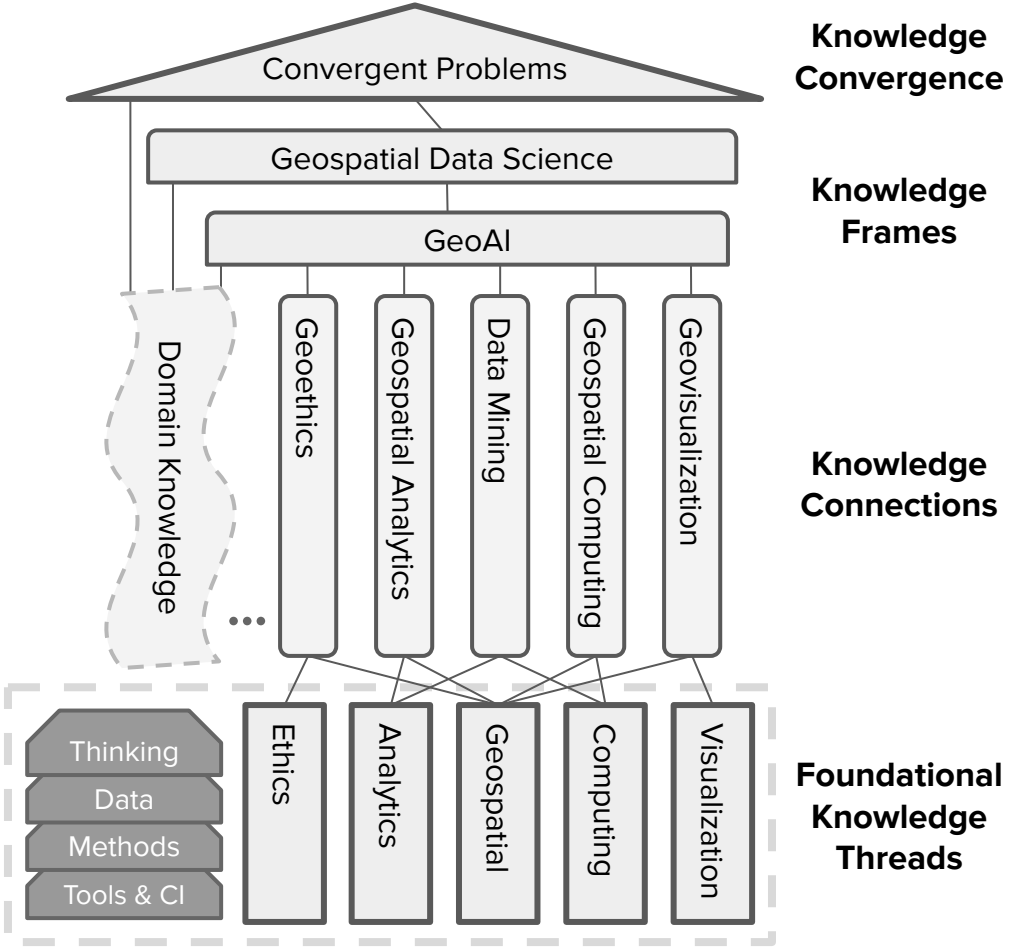
Check it out!



Multiple learner pathways

Technological Frame

Conceptual & Theoretical Foundation



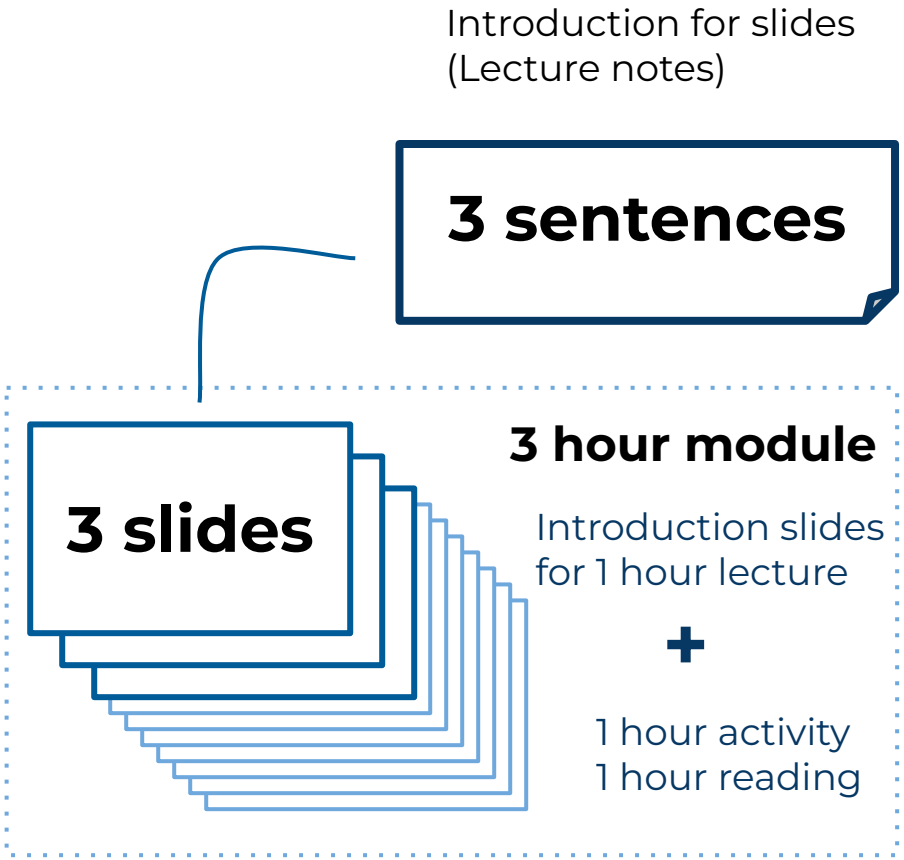
Modular Structure

Levels	**Time**	**Organization**	**Bloom's Tax.** ***(Verb)***	**Example LO**
3 sentences	**30 seconds**	1) Definition 2) Key idea 3) Example	Remember *(Define)*	Define GeoAI
3 slides	**3 minutes**	1) Definition 2) Example 3) Question/Discussion	Understand *(Recognize)*	Recognize one example of GeoAI
3 hour module	**3 hours**	1) Lecture (1h) (+ lecture notes) 2) Hands-on activity (1h) 3) Reading & reflection (1h)	Apply *(Use)*	Use GeoAI in an example problem

https://iguide.illinois.edu/convergence-curriculum-for-geospatial-data-science

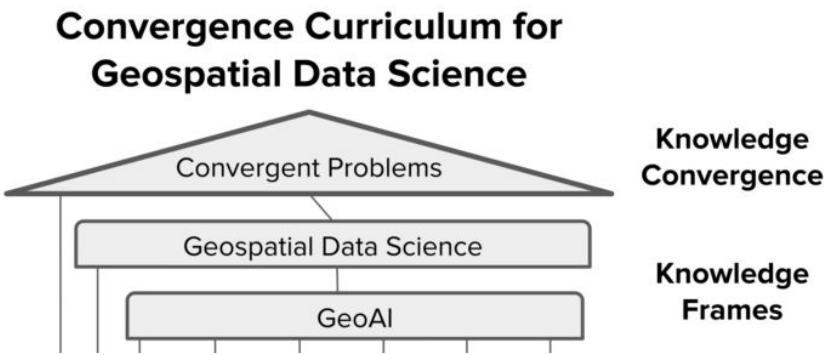
Modular Structure

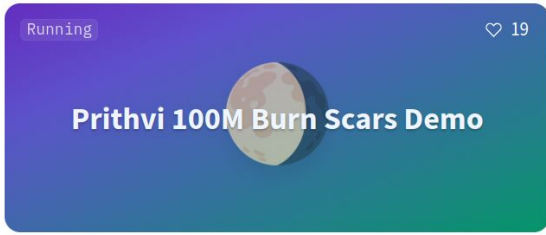
- Structure components to be used in multiple settings
- Organized production and review process
- Use Open Education Resources (OERs) when possible



Team 6 (Team Awesome @ Summer School)

Explored the geospatial artificial intelligence (GeoAI) foundation model by NASA and IBM called Prithvi-100M, which leverages satellite imagery data to provide insights into various geographical applications.





https://i-guide.io/i-guide-vco/introduction-to-the-geoai-foundation-model-and-its-application/

UCGIS Community Champions

- Bridging Geography with Language Models
- Exploring Street-View Imagery with Computer Vision
- Democratizing Spatial AI for Scalable Urban Tree Canopy Mapping
- Spatially-explicit Species Distribution Modeling and Prediction with Maxent
- Empowering GeoAI Education through Open-Source Spatial AI Workflows

Bridging Geography with Language Models

Yao-Yi Chang, University of Minnesota

This hands-on lab introduces students to SpaBERT, a spatially informed language model that enhances transformer architectures with geospatial pretraining objectives. The module will guide learners through training and applying SpaBERT to geographic datasets, helping them understand how spatial proximity and semantic context combine to improve the representation of geographic entities. Students work with global and regional datasets (e.g., OpenStreetMap features or gazetteers) through workflows to preprocess geographic data, integrate spatial embeddings, and generate region-level embeddings that capture semantic, spatial, and topological patterns. Emphasis is placed on constructing and visualizing contextual region representations to support downstream tasks like population prediction or movement analysis. The lab will have conceptual and practical components to prepare students for applying spatial language models in domains like urban analytics, transportation, and geographic information science. It equips students with skills in spatial AI, natural language processing, and data-driven geographic reasoning.

Exploring Street-View Imagery with Computer Vision

Alexander Michels, University of Texas at Dallas

Computer vision is an increasingly useful tool for geospatial problem-solving due to the proliferation of street-view imagery (SVI). SVI is well-known for its utility for self-driving cars as well as problems in safety, mobility, planning, economics and more. My spatial AI lab will cover the fundamental concepts in computer vision, open data sources for SVI data, and example analyses with an AI-ready dataset. Aimed at upper-level undergraduates and graduate students, the lab begins with foundational computer vision concepts, followed by an overview of SVI applications in safety, planning, and economics. Students will use an AI-ready dataset and apply pre-trained machine learning models (e.g., from Hugging Face) to perform spatial AI tasks. The lab concludes with links to additional learning resources from the I-GUIDE Platform and UCGIS GIS&T Body of Knowledge. It will be piloted in three courses between 2026 and 2027 and will include slides and a lesson plan for instructional use across disciplines.

Democratizing Spatial AI for Scalable Urban Tree Canopy Mapping

Yi Qi, University of Southern California

Urban trees provide critical ecosystem services that enhance environmental quality, community resilience, and human well-being. Traditional field-based tree inventories are costly and time-consuming, and recent advances in remote sensing and deep learning have opened new opportunities for mapping urban tree canopy at scale. This project will develop an open and transferable deep learning workflow for mapping urban tree canopy using freely available National Agriculture Imagery Program data. Building on prior work supported by the Bezos Earth Fund and piloted in Los Angeles, I will disseminate datasets, models, and training materials through the I-GUIDE Platform. By providing transparent workflows, instructional materials, and accessible tools, this project will not only deliver valuable resources for educators and students but also help democratize spatial AI for urban forestry, enabling local governments, researchers, and community organizations to monitor, plan, and sustain their tree canopy resources more effectively.

Spatially-explicit Species Distribution Modeling and Prediction with Maxent

Siqin (Sisi) Wang, University of Southern California

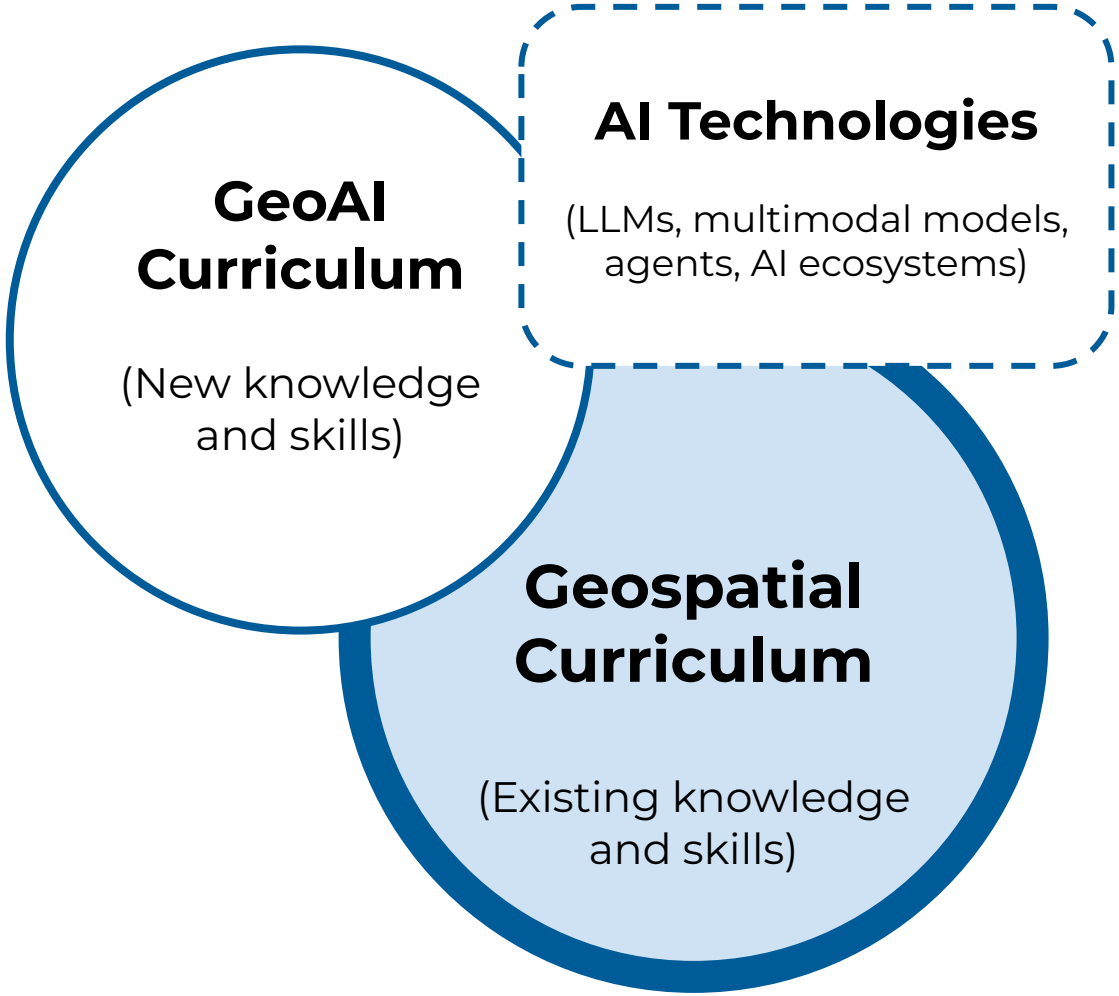
This project brings spatially explicit species distribution modeling (SOM) using Maxent (Maximum Entropy Modeling) into the classroom. Designed for advanced graduate-level spatial analysis courses, it integrates machine learning principles, spatial data manipulation, and model interpretation to teach students how environmental variables influence spatial distributions of particular species. Using real-world crop data from Hawaii, students combine georeferenced occurrence points with environmental and soil variables to predict suitable cultivation areas. The curriculum covers data preparation, model configuration, execution, and validation, emphasizing machine learning concepts and spatial interpretability. Students learn to adjust Maxent parameters, interpret ROC curves and AUC scores, and use jackknife tests and response curves to assess variable importance. The outputs can readily be visualized with GIS to enhance communication and spatial storytelling.

Empowering GeoAI Education through Open-Source Spatial AI Workflows

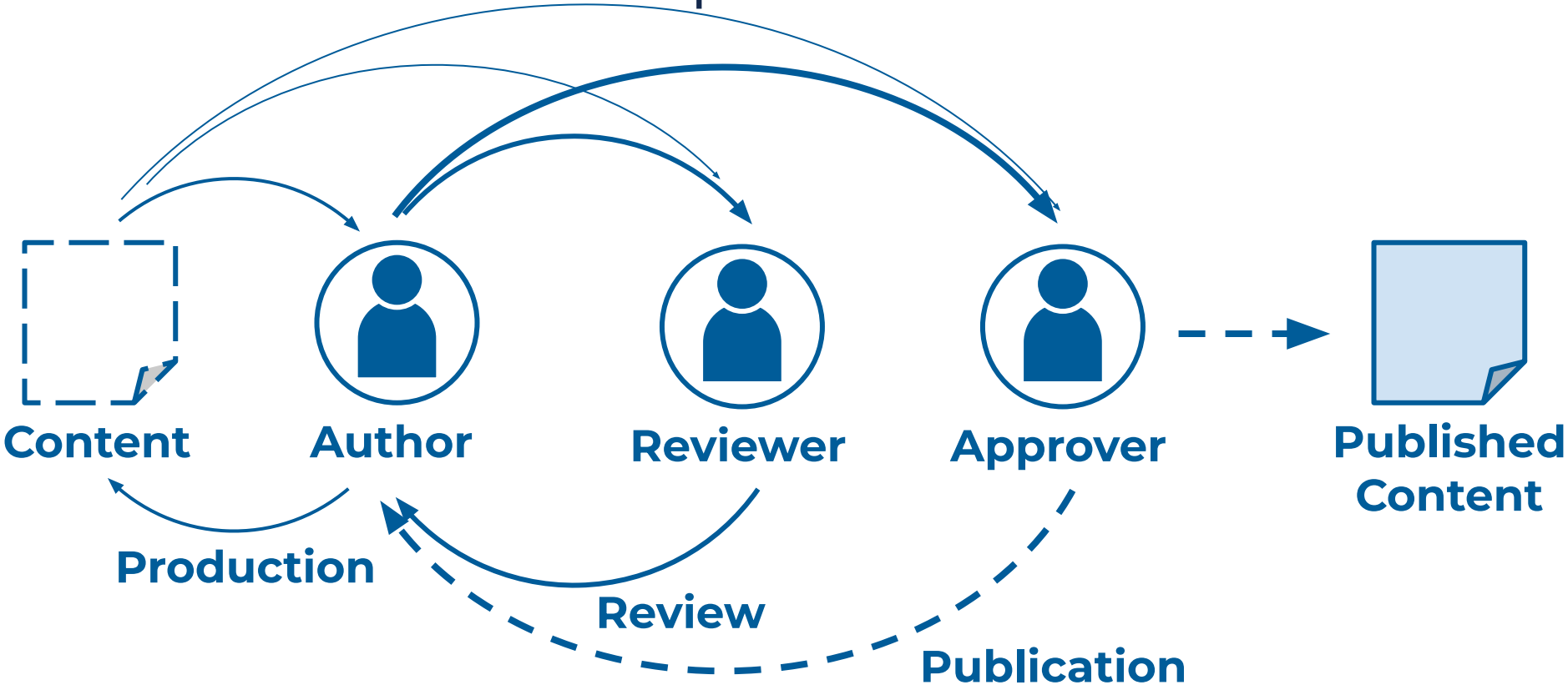
Qiusheng Wu, University of Tennessee at Knoxville

Spatial Artificial Intelligence (GeoAI) is transforming Earth system analysis, yet integrating these powerful tools into geospatial education is often hindered by accessibility and infrastructure challenges. This project addresses these barriers by developing open-source GeoAI teaching modules that combine cloud-based data access, interactive visualization, and hands-on deep learning for spatial analysis. Using Python tools like SAMGeo and GeoAI, learners will be guided through end-to-end workflows—including data acquisition from platforms like the Microsoft Planetary Computer and AWS Open Data, model training, and geospatial image segmentation. Designed for reproducibility and scalability, the modules will be hosted on I-GUIDE's JupyterHub as interactive notebooks. To broaden impact, the modules will be piloted through live training sessions and YouTube tutorials, and the I-GUIDE learning platform.

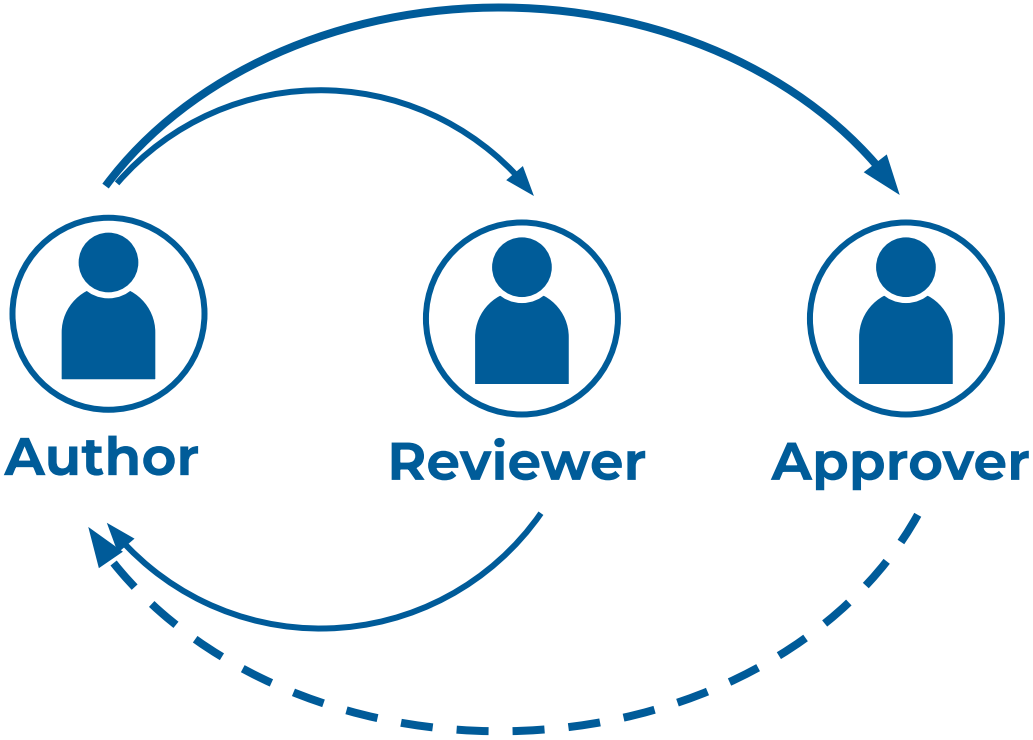
Curriculum Building + AI


AI Technologies
(LLMs, multimodal models, agents, AI ecosystems)
GeoAI Curriculum
(New knowledge and skills)
Geospatial Curriculum
(Existing knowledge and skills)

Curriculum Development - Process

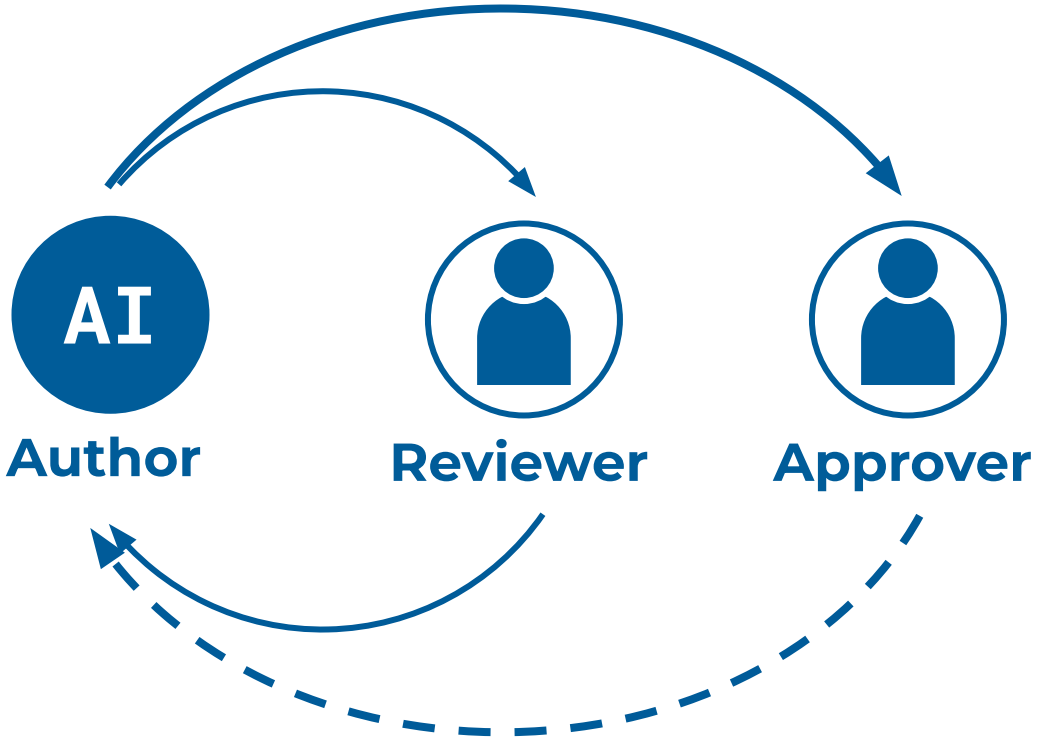


Curriculum Development - Human Loop



I-GUIDE VCO
March 11, 2026

Human-in-the-Loop (Creator)



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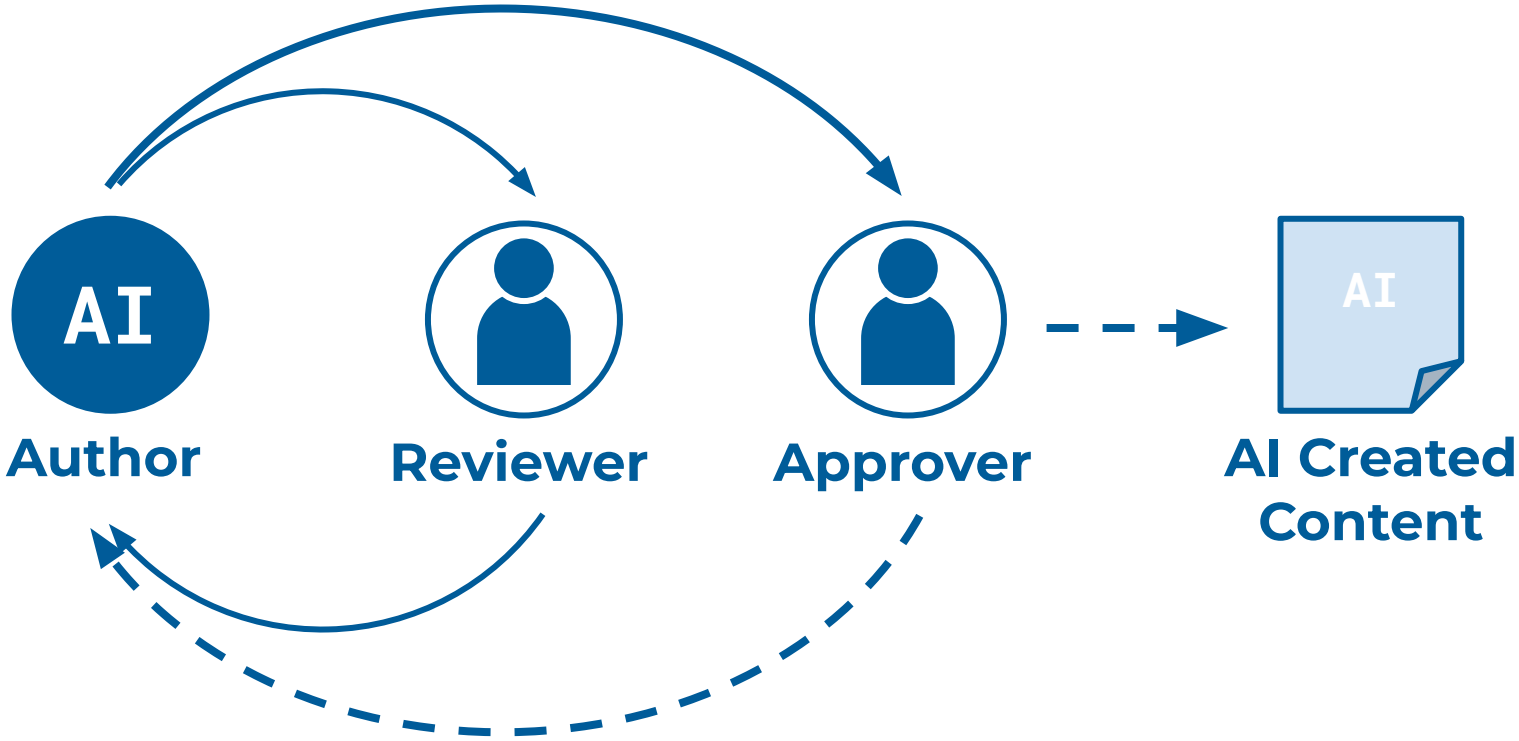

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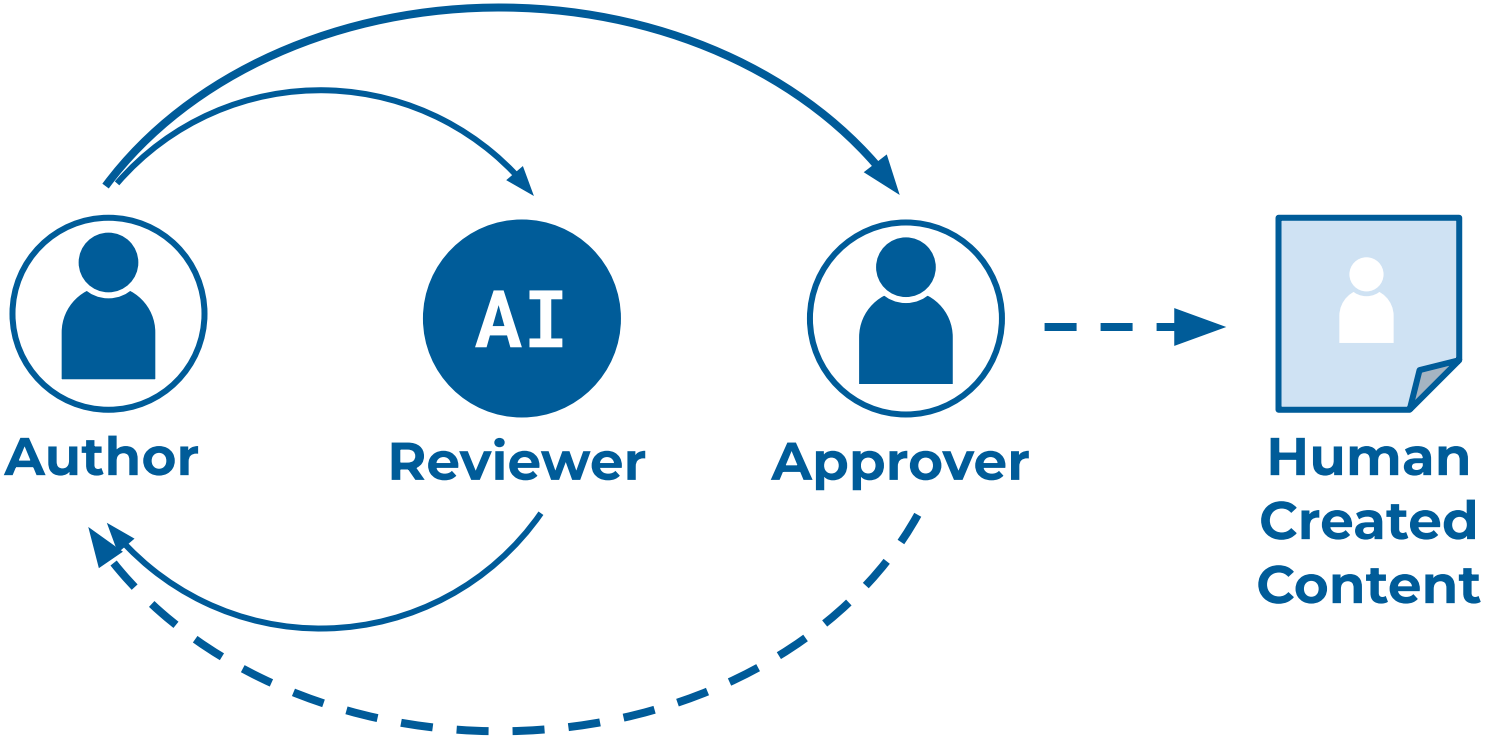
Human-in-the-Loop (Reviewer)



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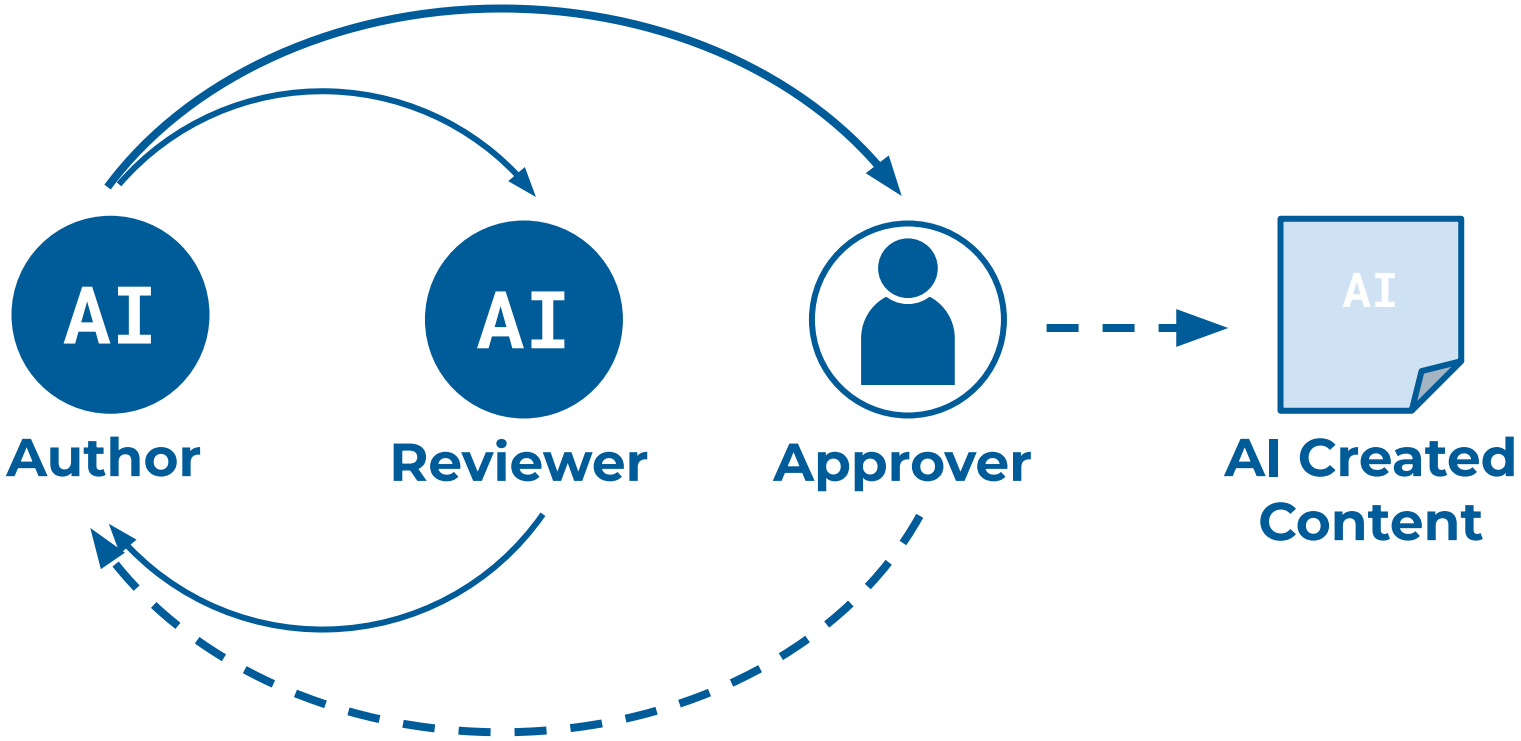
Human-in-the-Loop (Pre-Publication)



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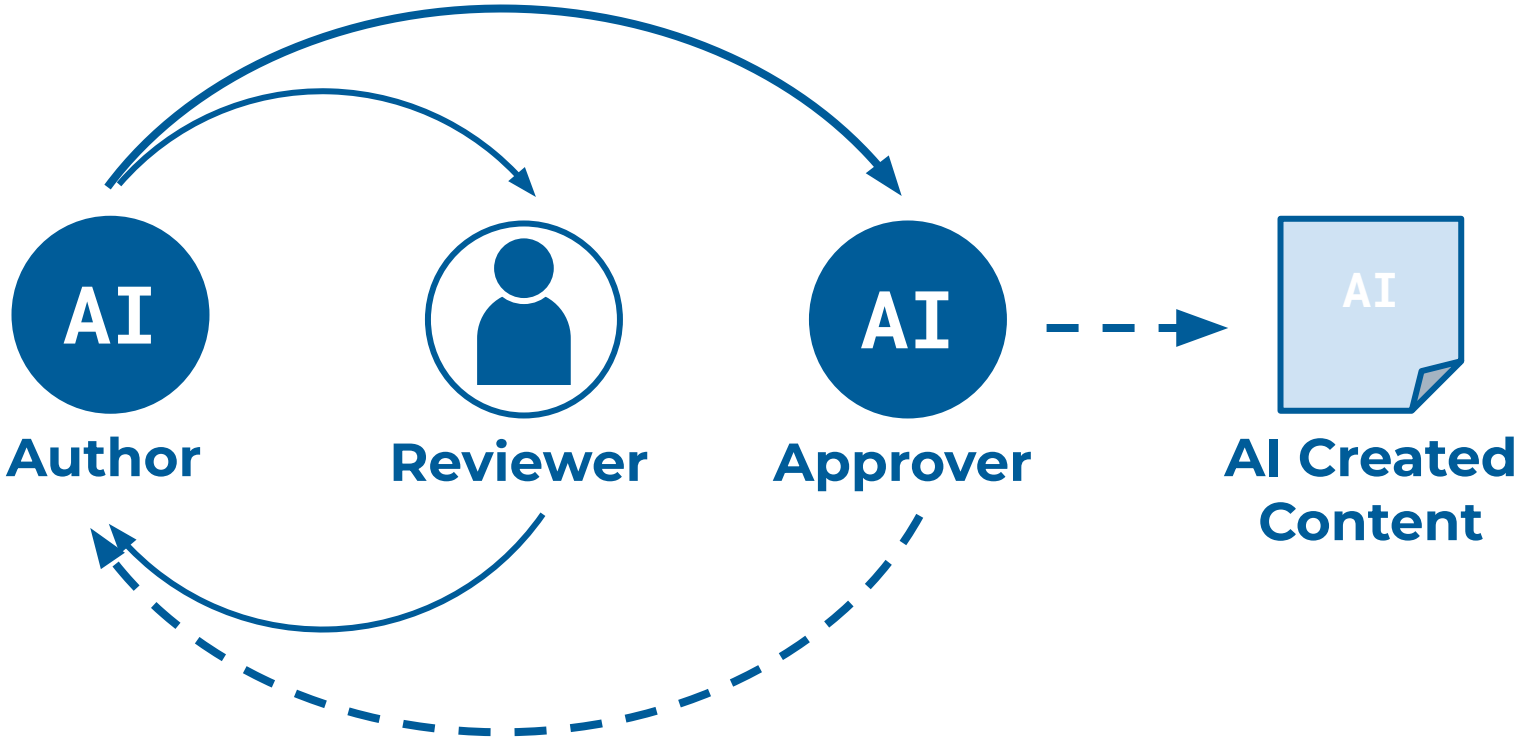
Human-on-the-Loop (Overseen)



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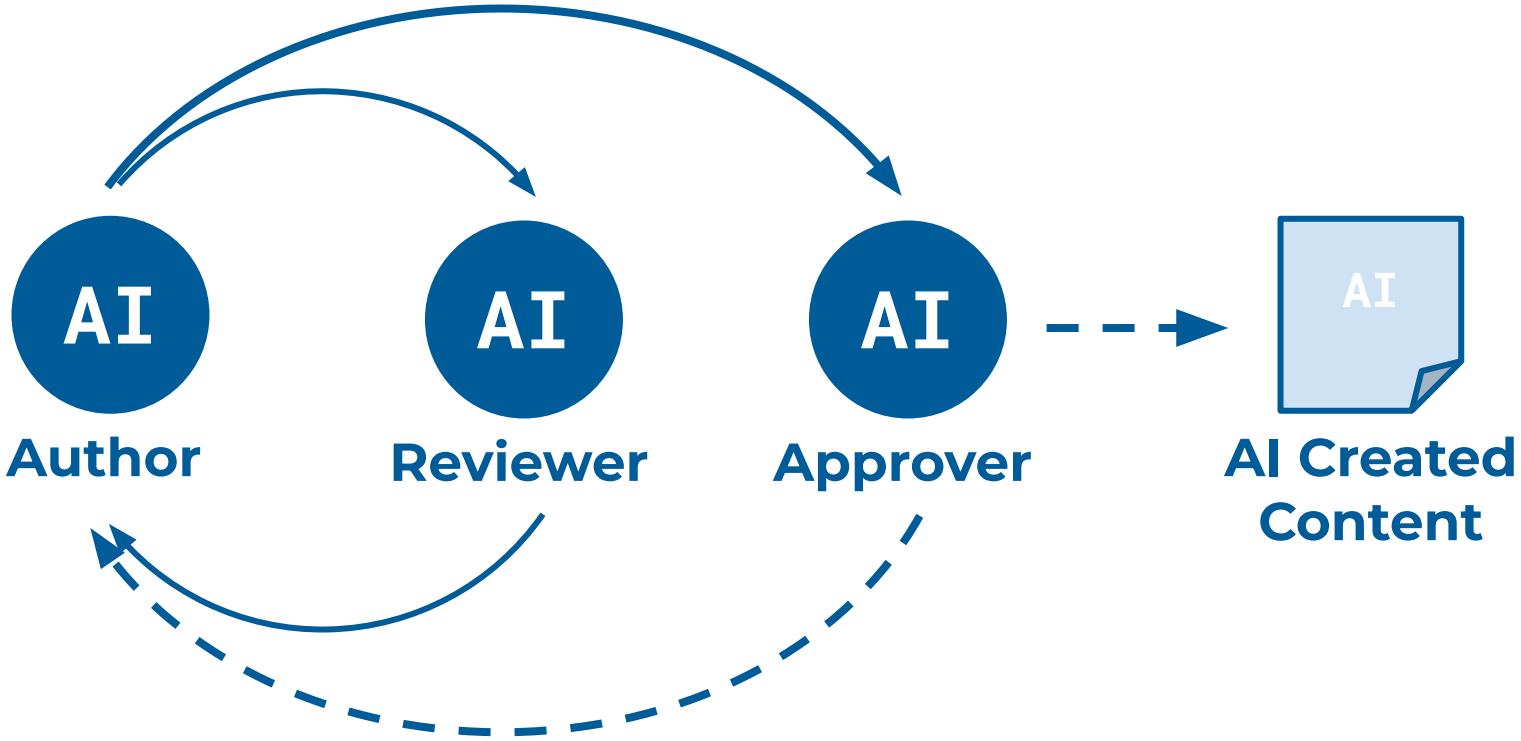
Human-out-of-the-Loop (Autonomous)



I-GUIDE VCO
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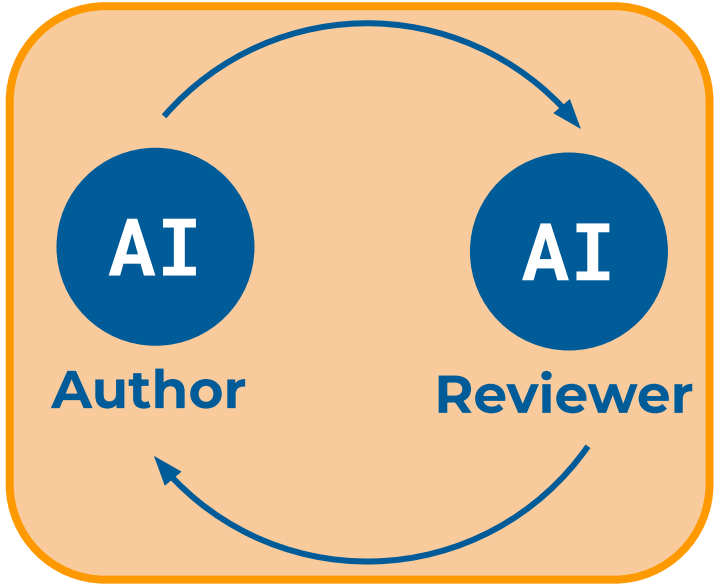
Where are we going with Agentic Curriculum Building

Our Focus: Agentic Development



AI Interface Prototype (Jinzhe Wang)

Enter topics (comma-separated, up to three) to generate presentations and videos.

Topics

Machine Learning

Upload PPT Template (Optional)

template.pptx 43.5 KB

Generate PPT & Video

Preview

Topic 1:
Title: Machine Learning
Description: Machine Learning is a subset of artificial intelligence that focuses on the development of algorithms that allow computers to learn from and make predictions based on data. It involves training models on datasets so they can recognize patterns and improve their performance over time. With applications ranging from image recognition to natural language processing, machine learning is transforming various industries.
Discussion: How can machine learning be utilized to improve decision-making processes in businesses?

Downloads

PPT Files

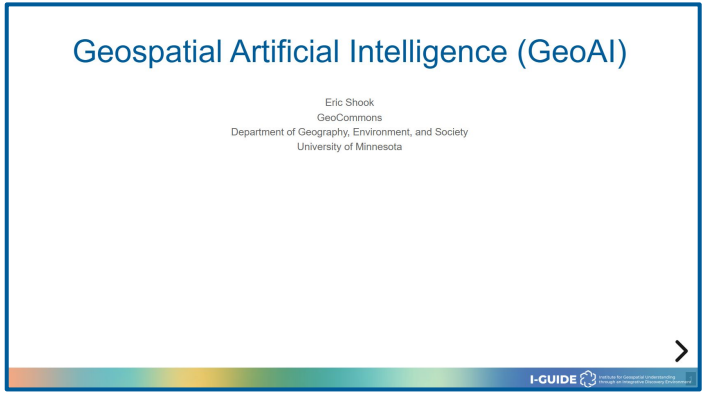
Machine_Learning_presentation.pptx 42.3 KB

Video Files

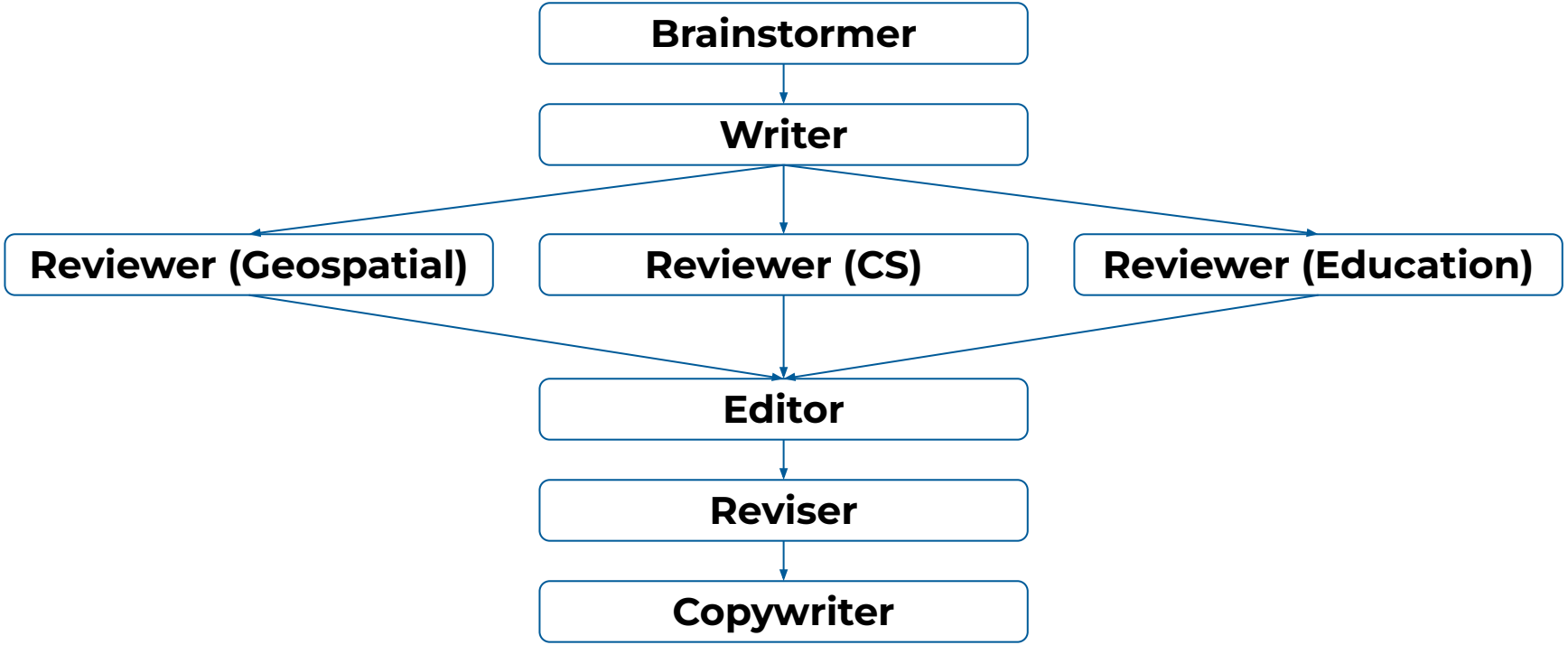
Machine_Learning_presentation.mp4 1.1 MB

AI Workflow Prototype (Jinzhe Wang)

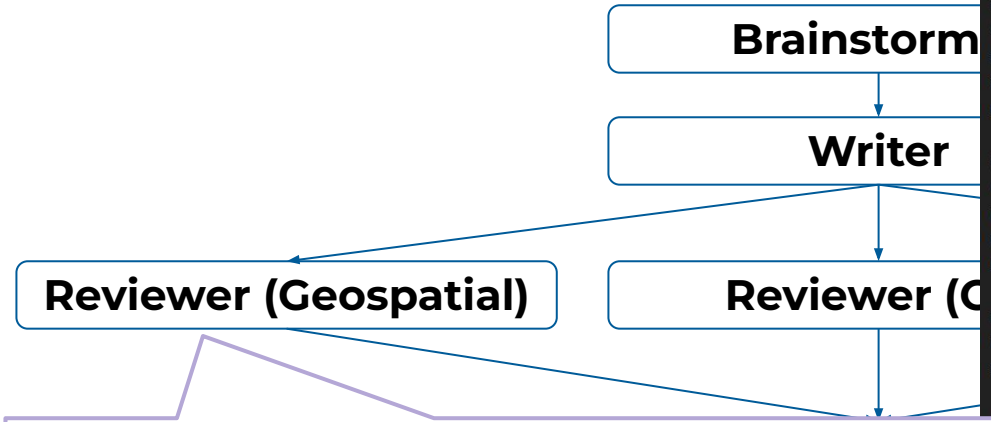
User Input → Agentic Workflow → Content Generating → Slides Generation → Audio Generation → Video Output



Agentic Workflow Builder (Base)

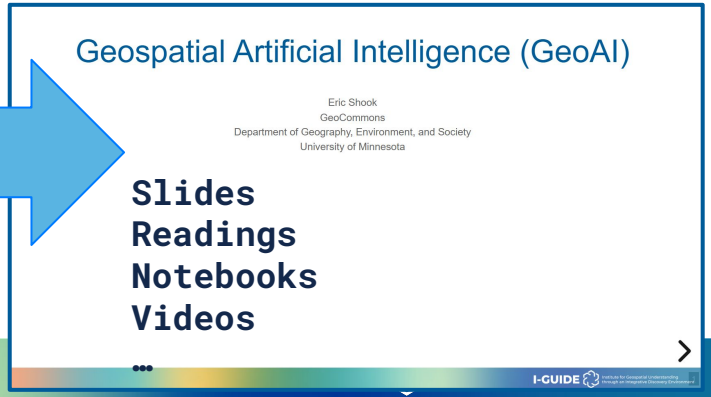


Agentic Workflow Buil



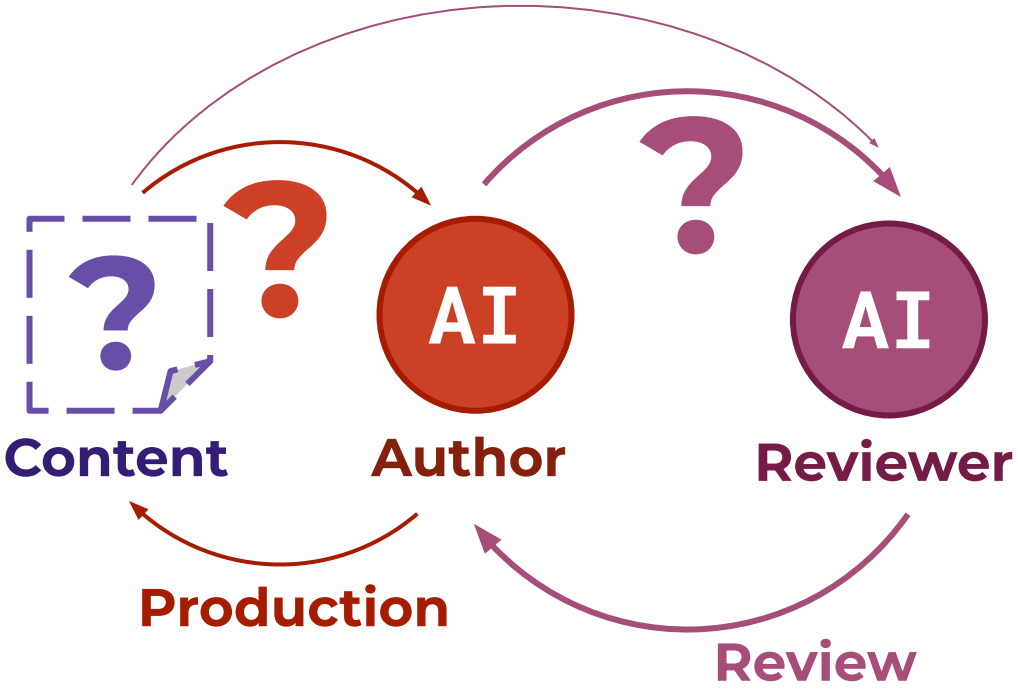

```python
# The start
workflow.set_entry_point("brainstormer")

workflow.add_edge("brainstormer", "writer")

# Multiple reviewers
# FAN-OUT: Writer goes to all reviewers at once
workflow.add_edge("writer", "reviewer_a")
workflow.add_edge("writer", "reviewer_b")
workflow.add_edge("writer", "reviewer_c")

# FAN-IN: All reviewers point to the Editor
# LangGraph automatically waits for ALL THREE to finish before starting the Editor.
workflow.add_edge("reviewer_a", "editor")
workflow.add_edge("reviewer_b", "editor")
workflow.add_edge("reviewer_c", "editor")

# Bring them all back and revise
workflow.add_edge("editor", "reviser")
workflow.add_edge("reviser", "copywriter")

# The end
flow.add_edge("copywriter", END)
```

"You are a skeptical academic from GIS and Geography. Review this for geographic theory gaps and misunderstanding of GIS concepts and technologies"



Code here: ttps://platform.i-guide.io/notebooks/2255396f-2acd-4341-9eda-0e27cd4cdc43

Agentic Curriculum Experimentation



1. Content **Type**
2. Content **Length**
3. Review **Process**
4. Production **Process**

Q & A

Acknowledgements

I-GUIDE support, National Science Foundation (NSF) (OAC-2118329)

Hour of CI support, National Science Foundation (NSF) (OAC-1829708)

NSF-supported XSEDE and XSEDE-ECSS (ACI-1548562)

Hour of CI project contributors, partners, lesson developers, pilot instructors, and students

I-GUIDE team

I-GUIDE Institute for Geospatial Understanding through an Integrative Discovery Environment

Thank you!

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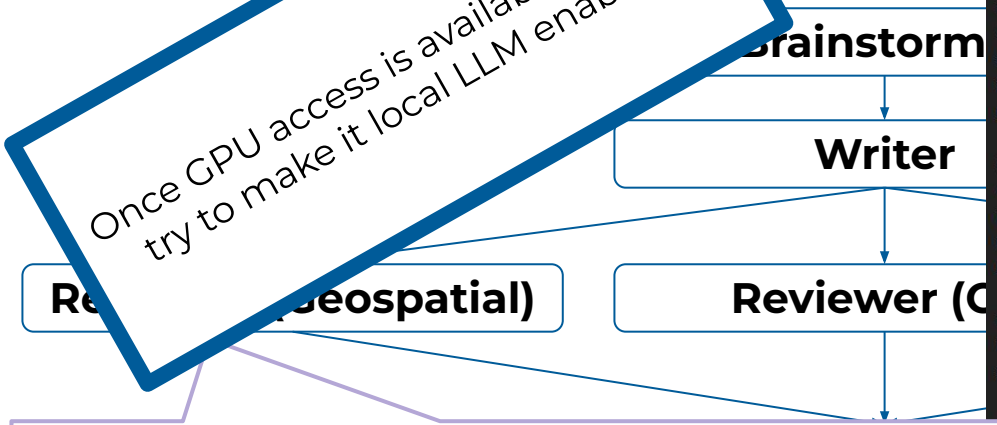
Agentic W w Buil

Once GPU access is available, we will try to make it local LLM enabled.

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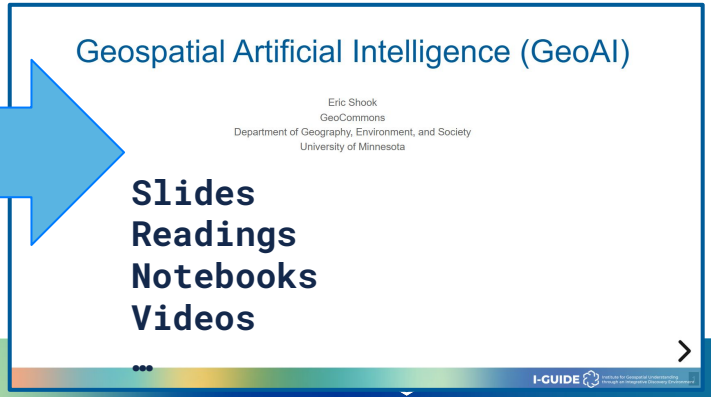

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